# Link Grades

Epistemic and Classical Planning over a Link
that Is Never, Partly or Always Down

Haniel Ulises

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Two robots must both come to know whether a corridor is blocked. Only a robot at the corridor's mouth can decide it with its laser, each robot builds its own occupancy map, and what one map holds reaches the other robot only over a link that a monitor takes down never, for a window of the mission, or for all of it. We compare five formulations on the same seeds: two epistemic ones, planned by Aletheia over one EPDDL domain in which a message is delivered or lost, and three classical ones, planned by POPF. We show that a planner that minimises the depth of the worst branch, without costs or likelihoods, never plans to try a lossy link and fall back when the link's state is unknown: the branch in which the message is lost is one step deeper than the fallback alone. With the link unknown, Aletheia's policy therefore sends both robots to look, and it coincides with the classical plan that omits the link; with the link read before planning and given as common knowledge, its policy shares and recovers by replanning after a timeout, and it coincides with the classical report and replan. In Gazebo, over 20 seeds, three grades and the five formulations, and judging each run by whether each robot's own map decides the corridor at its end, the policy with the link unknown met the goal in 60 of 60 runs in 62.7 s on average, and the fire-and-forget classical report, reported successful in every run, was backed by the receiver's map in 20 of 60.

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## 1 Scope

This report accompanies the page *Link grades* of the Epistemic Robotics project and the package `link_grades_demo`. It states the environment and the domain, proves the property of worst-case planning that decides what the epistemic planner does with a lossy link, describes the five formulations, and reports a batch of 300 runs in Gazebo. The experiment is the repository's counterpart of a comparison in the author's thesis, where it is treated separately.

## 2 Environment

The hall is 14 by 8 m. Along its north side runs a corridor 5 m long and 1 m wide, a dead end whose mouth opens west. In half the seeds a box of debris stands in its middle. From anywhere but the mouth the corridor's walls hide it, and a ray down the corridor ends on its end wall, so that a corridor without debris is seen to be empty and not merely not seen. The robot $r1$ starts at the west dock, about 6 m of route from the mouth; $r2$ at the east station, about 13 m. Each has a spot at the mouth, side by side (Figure 1).

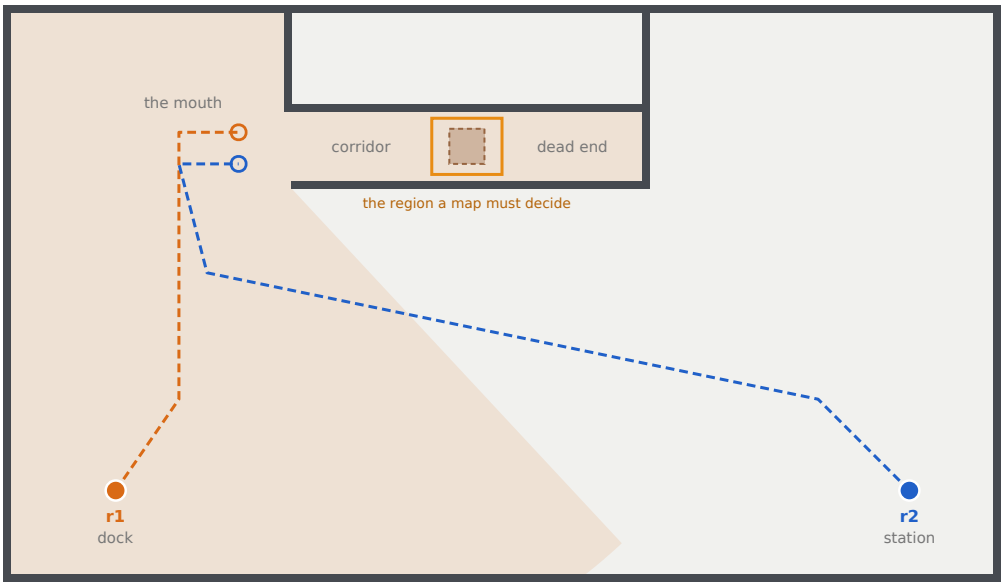



**Figure 1:** The hall. Shaded: what a laser reaches from $r1$'s spot with the corridor empty. Dashed box: the debris. Orange outline: the region a map must decide. Dashed lines: each robot's route.

Each robot integrates its own scans into an occupancy grid of 5 cm cells. A grid decides the corridor when its cells in the outlined region say so: blocked when at least 6 of them are known occupied, clear when none is and at least 90 % of its 10 cm blocks hold a known free cell; otherwise it does not decide. Over the link each robot also receives the other's grid and merges it under its own, its own cell where it has one and the other's where it has none. What a robot knows about the corridor is read off this merged map.

The link is cut by the `comm_monitor` and `link_gate` of the project's link-outage demonstration. The monitor declares the link up or down on a fixed schedule in simulated time, and a gate on each side passes or drops every message between the robots accordingly, grids, decisions, reports and acknowledgements, in both directions at once. Under the *full* grade the link is never down; under the *partial* grade it falls 10 s into the mission and returns 24 to 100 s later, the outage growing by 4 s with each seed; under the *none* grade it is down before the mission starts and never returns.

## 3 Domain

The domain `link-grades.epddl` has three atoms, *blocked*, *at-junction*$(i)$ and *link-up*, and three actions, given as event models of dynamic epistemic logic [1] and planned over as in [2]. The action goto-junction$(i)$ is public and ontic. The action inspect$(i)$ is semi-private sensing with two designated events, one per value of *blocked*, each requiring that $i$ stand at the junction; $i$ observes which occurs and the other robot observes only that it occurred. The action share$(r1, r2)$ has the action type `lossy-message` of the library `link.epddl`: three designated events with trivial postconditions, delivered with *blocked*, delivered with $\neg$*blocked*, and lost, the first two requiring *link-up* and the third $\neg$*link-up*, every one observed fully by both robots. The receiver observes it because the message arrives or does not, the sender because the acknowledgement, carried by the same link, does or does not. The goal is

$$Kw_{r1}\, \mathit{blocked} \;\wedge\; Kw_{r2}\, \mathit{blocked},$$

which does not mention the link. Three initial models are used: `problem_open`, in which neither robot knows *blocked* or *link-up* (four worlds, all designated); `problem_up` and `problem_down`, in which *link-up* or $\neg$*link-up* is common knowledge; and `problem_lost`, with $r1$ at the junction knowing whether *blocked*, $r2$ not knowing it, and the link down.

## 4 Worst-case planning and a lossy message

Aletheia's AO* [3] returns a policy whose worst branch is of least depth. The following property decides what it does with the share.

**Proposition 1.** Let $s$ be an epistemic state in which *link-up* is not known to be true by the agents, and suppose that (i) neither the goal nor the precondition of any action other than share mentions *link-up*, (ii) the lost event of share has trivial postconditions, (iii) share is applicable in $s$, and (iv) *link-up* is independent of every other atom in $s$: for each world there is one that differs from it only in *link-up*, and no agent can tell the two apart. Let $d^*(s)$ be the least worst-case depth of a policy that reaches the goal from $s$, and $d^*_{-}(s)$ the least such depth among policies that never apply share. Then every policy that applies share in $s$ has worst-case depth at least $1 + d^*_{-}(s)$, and therefore $d^*(s) = d^*_{-}(s)$ and no optimal policy applies share in $s$.

*Proof.* Because *link-up* is not known true in $s$, some designated world satisfies $\neg$*link-up*, the lost event is applicable there, and a policy that applies share in $s$ has a branch $s' = s \otimes$ share restricted to the lost event. By (ii) the facts of every world of $s'$ are those of $s$; the lost event, observed by both agents, only removes the worlds in which *link-up* holds. By (iv), $s'$ and $s$ agree, up to bisimulation, on every formula that does not mention *link-up*; by (i), a policy that never applies share is then applicable, and reaches the goal, from $s$ exactly when it does from $s'$, branch for branch, since nothing it does or aims at depends on *link-up*. In $s'$ the link is known to be down, so share has only its lost event there and returns $s'$ itself; an optimal policy from $s'$ therefore never applies it, and $d^*(s') = d^*_{-}(s') = d^*_{-}(s)$. The branch through $s'$ has depth $1 + d^*(s') = 1 + d^*_{-}(s)$, which exceeds the depth $d^*_{-}(s)$ of the best policy without share. □

The conditions hold in `problem_open` at every state in which $r1$ could share: its four worlds are the product of the values of *blocked* and *link-up*, and only inspecting tells the robots anything about *blocked*. On that task Aletheia returns, at depth 4 after 23 expansions, the policy

```
goto-junction(r1) · inspect(r1)
   on e-inspect-blocked or e-inspect-clear:  goto-junction(r2) · inspect(r2)
```

in which both robots go and look. This is the formulation *epistemic, link unknown*. With *link-up* common knowledge and true, the hypothesis of the proposition fails, the lost event is not applicable, and on `problem_up` Aletheia returns, at depth 3,

```
goto-junction(r1) · inspect(r1)
   on e-inspect-blocked or e-inspect-clear:  share(r1, r2)
```

The mission of the formulation *epistemic, link read* asks the monitor for the link's state before planning and plans on `problem_up` or `problem_down`. When the link falls after the reading, the share is not acknowledged within 30 s, the executor reports the action failed, and the mission loads `problem_lost` and replans, which sends $r2$ to look. A policy that tries the link and falls back within the plan would require a criterion that weighs the delivered branch against the lost one, such as an expected cost or a bound on the probability of loss; Aletheia has neither, and neither the domain nor the planner was altered to obtain one.

## 5 Classical planning

The classical domain has the same drives and inspections, a fact $knows(r)$ for each robot and a fact *link*. A report from $r1$ to $r2$ requires *link* and makes $knows(r2)$ true, and the goal is $knows(r1) \wedge knows(r2)$. Three formulations differ in how the link enters the problem. In *visit* the problem omits *link*, and POPF [4], through PlanSys2 [5], sends both robots to look. In *report* the problem always includes *link* and the report is sent without waiting for anything. In *report and replan* the problem includes *link* only when the monitor says the link is up; the report then waits up to 30 s for an acknowledgement and fails without one, and the mission removes *link* from the problem expert's state and replans. The wait of the report and the wait of the epistemic share are one parameter in one node. None of the three states the goal as knowledge: the fact $knows(r2)$ holds in the plan once a report is applied, and only the acknowledgement logic of the third formulation, outside the classical model, keeps it from holding when the message was lost.

By construction the unknown link coincides with the visit and the read link with the report and replan. The comparison on time therefore measures the assumptions about the link; what separates the epistemic formulations from their classical counterparts is that their goal is a knowledge formula the model checks, and that the open formulation reaches the robust policy from a model that represents the link and the robots' ignorance of it.

## 6 Measurement

A run is *reported* when the executor ends the mission with success. It is *backed* when, in addition, each robot's merged map decides the corridor correctly at the end, and *unbacked* when it is reported and not backed. Mission time is measured in simulated seconds from a common start, at 25 s of simulated time. The time at which the later of the two maps first decided the corridor correctly is recorded separately, whether or not the mission had ended, together with the distance each robot drove and the area of the hall its map holds out of the 101.8 $m^2$ a robot can see.

## 7 Evaluation over 20 seeds and three grades

Table 1 gives the outcome per formulation and grade, and Table 2 the coverage and the distances.

**The two epistemic formulations coincide with their classical counterparts.** Paired by seed and grade, the read link and the report and replan differ in mission time by 0.3 s in the median and 1.2 s at most; the unknown link and the visit by 0.3 s and 0.8 s. The policy with the link unknown takes 62.7, 62.6 and 62.7 s under the three grades: r2's drive across the hall sets its time, and the link does not enter it. This is also its cost. Under the full grade r2's merged map decided the corridor 22.9 s into the mission, from the grid r1 sent, and the policy, which does not count on the link, still sent r2 to the mouth.

Reading the link before planning pays when the link holds: under the full grade the read link finishes in 30.5 s, since r2 never drives. Under the partial grade the outage outlasted the acknowledgement's wait in 13 of 20 seeds; the share failed, the mission replanned, and r2 drove

| formulation | grade | backed | reported | unbacked | replanned | time (s) | decided (s) |
|---|---|---|---|---|---|---|---|
| epistemic, link unknown | full | 20/20 | 20/20 | 0 | 0 | 62.7 | 22.9 |
| | partial | 20/20 | 20/20 | 0 | 0 | 62.6 | 53.2 |
| | none | 20/20 | 20/20 | 0 | 0 | 62.7 | 56.6 |
| epistemic, link read | full | 20/20 | 20/20 | 0 | 0 | 30.5 | 22.9 |
| | partial | 20/20 | 20/20 | 0 | 13 | 96.6 | 73.3 |
| | none | 20/20 | 20/20 | 0 | 0 | 62.7 | 56.7 |
| classical, visit | full | 20/20 | 20/20 | 0 | 0 | 62.7 | 23.2 |
| | partial | 20/20 | 20/20 | 0 | 0 | 62.7 | 53.4 |
| | none | 20/20 | 20/20 | 0 | 0 | 62.6 | 56.8 |
| classical, report | full | 20/20 | 20/20 | 0 | 0 | 30.6 | 23.1 |
| | partial | 0/20 | 20/20 | 20 | 0 | 30.7 | never |
| | none | 0/20 | 20/20 | 20 | 0 | 30.6 | never |
| classical, report and replan | full | 20/20 | 20/20 | 0 | 0 | 30.6 | 23.2 |
| | partial | 20/20 | 20/20 | 0 | 13 | 96.8 | 73.3 |
| | none | 20/20 | 20/20 | 0 | 0 | 62.8 | 56.9 |

**Table 1:** Runs per cell out of 20. Backed: reported, and each robotś map decides the corridor correctly at the end. Time: mean mission time over reported runs. Decided: mean time at which the later map first decided the corridor correctly.

only after the wait, so that these runs took 123.2 s on average. In 26 of the 26 replanned runs of the read link and the report and replan, the link returned while r2 was still driving, and r2's merged map decided the corridor from r1's cells 30.3 s on average before r2 reached the mouth; the replanned plan had no step that used it. Under the none grade the reading finds the link down, and both formulations plan as the visit does.

The classical report was reported successful in every run. Under the full grade its success was backed, by the grid r1 sends continuously and not by the report itself; under the partial and none grades the report left while the link was down, and 40 of its 40 runs there ended with r2's map deciding nothing. No map, in any run, decided the corridor wrongly.

| formulation | grade | r2 own ($m^2$) | r2 merged ($m^2$) | r2 drove (m) | both drove (m) |
|---|---|---|---|---|---|
| epistemic, link unknown | full | 101.0 | 101.2 | 12.8 | 18.8 |
| | partial | 101.0 | 101.1 | 12.8 | 18.8 |
| | none | 101.0 | 101.0 | 12.8 | 18.8 |
| epistemic, link read | full | 42.4 | 94.9 | 0.0 | 6.0 |
| | partial | 80.6 | 97.6 | 8.3 | 14.3 |
| | none | 101.0 | 101.0 | 12.8 | 18.8 |
| classical, visit | full | 101.0 | 101.2 | 12.8 | 18.8 |
| | partial | 101.0 | 101.1 | 12.8 | 18.8 |
| | none | 101.0 | 101.0 | 12.8 | 18.8 |
| classical, report | full | 42.4 | 94.9 | 0.0 | 5.9 |
| | partial | 42.4 | 90.9 | 0.0 | 6.0 |
| | none | 42.4 | 42.4 | 0.0 | 6.0 |
| classical, report and replan | full | 42.4 | 94.9 | 0.0 | 5.9 |
| | partial | 80.6 | 97.8 | 8.3 | 14.3 |
| | none | 101.0 | 101.0 | 12.8 | 18.8 |

**Table 2:** Area of the hall held by r2ś own and merged maps at the end, and distance driven, as means over the cellś runs.

Coverage follows the drives. Where r2 drives to the mouth, its own map holds 101.0 $m^2$ at the

end; where it stays at the station, 42.4 $m^2$. Over a link that holds, the merged map makes up the difference: 94.9 $m^2$ for the read link under the full grade, against 101.2 $m^2$ for the policy that sends r2. The region that decides the goal is in both maps in every backed run; the area beyond it measures the driving, and the price of the drive is the time in the table above.

## 8 Recorded runs

The film on the page shows four runs on seed 15, in which the corridor is blocked and the partial outage lasts 80 s: the policy with the link unknown under the partial grade, the read link under the full grade and under the partial grade, and the classical report under the partial grade. Figure 2 is a frame of the third.

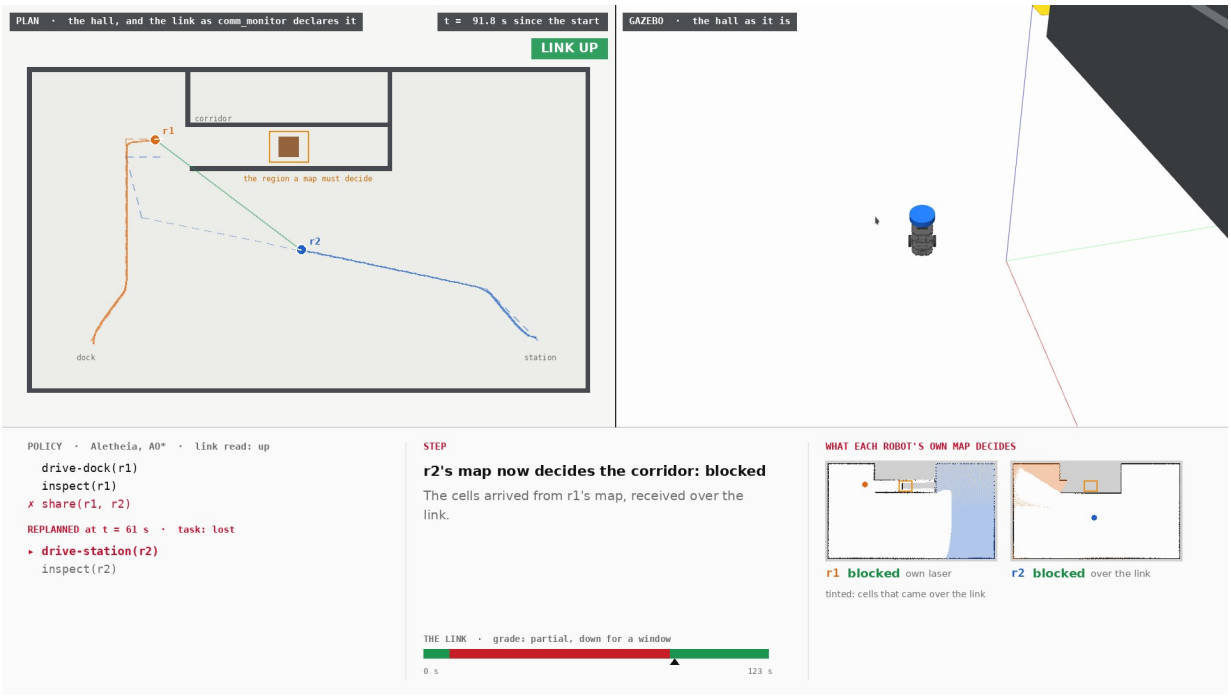



**Figure 2:** The read link under the partial grade, as $r2$'s merged map decides the corridor from $r1$'s cells after the link has returned, while $r2$ is still driving to look.

## 9 Implementation defects

Two defects were found while the experiment was built. The first is ePlanSys's executor checking the modal conjuncts of all of an action's designated events together, as a single conjunction, so that a share whose delivered events required $K_{r1}$ *blocked* in one and $K_{r1}\neg$*blocked* in the other could never be dispatched. Each such event now requires $Kw_{r1}$ *blocked* together with its own fact, which is equivalent. The second concerned the classical report and replan: the mission removed *link* from the problem expert's state by matching its text, which the problem expert prints with inner spaces, so the replan kept the link and tried the report again. The match was corrected before every run reported here. One run, the unknown link under the partial grade on seed 4, was discarded and run again: its simulator's clock never advanced past the start, and the run reached the wall-time limit with both drives dispatched and neither robot moved.

## 10 Limitations

The results hold under several restrictions of the model. The link drops both directions together, so that a message cannot arrive while its acknowledgement is lost; with independent directions

the receiver could know while the sender does not know that it knows, and the share would need a fourth event or a weaker observability for the sender. Proposition 1 concerns planners that minimise the depth of the worst branch; under an expected cost, a policy that tries the link could be optimal when the link is likely up.

The experiment has restrictions of its own. The two epistemic formulations coincide by construction with two classical ones, so that time measures the assumptions about the link and not the planners. The partial grade cuts the link at a fixed instant for an outage whose length grows with the seed; in the 7 seeds whose outage ends within the acknowledgement's wait, a read link is acknowledged late and no replanning occurs. Mission time includes the planners' own wall time, during which the simulation advanced.

## References

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